

Microwave Control-Chain Spoofing of NV Magnetometers: Detection Limits and Reference-Sensor Design Rules

Sounak Bhowmik and Himanshu Thapliyal
Department of Electrical and Computer Engineering
Southern Methodist University, Dallas, TX, USA
sbhowmik@smu.edu, hthapliyal@smu.edu

Abstract—Diamond magnetometers based on nitrogen-vacancy centers are moving from the laboratory to field applications such as magnetic anomaly navigation, where a wrong field estimate becomes a wrong position. The quantum sensor is hard to reach remotely, but its classical microwave control chain is not. We show that an attacker who offsets the microwave drive frequency by less than one part per million reproduces the photon statistics of a genuine field shift for any Ramsey interrogation schedule, so the sensor cannot verify its own output. Secret interrogation timing stops precomputed phase attacks but not frequency offsets, and driving both spin transitions analytically reduces a clock error’s bias roughly tenfold. We then analyze the common defense of falling back to an auxiliary reference sensor on alarm. Under attack, the rate of harmful fallback outputs cannot fall below the reference’s own error rate, and a closed-form design equation predicts a blind window of small spoofs that pass unnoticed, which fixes how precise the reference must be. In a simulation study with 30 independently seeded pipelines and a protocol fixed before the run, the equation matches all eight tested reference conditions within one percentage point, and a reference that shares the primary sensor’s clock releases a harmful estimate on almost every attack.

Index Terms—NV magnetometry, quantum sensor security, sensor spoofing, microwave control, reference sensor fusion, Ramsey magnetometry

I. Introduction

Nitrogen-vacancy (NV) magnetometers combine high sensitivity with room-temperature operation, which makes them candidates for field deployment [1]. One such application is magnetic anomaly navigation, where the measured field is matched against a map to obtain position [2]. In this setting a field error is a navigation error, and the integrity of the released estimate matters as much as its precision.

An NV Ramsey magnetometer reads the field as a phase that accumulates between two microwave pulses. The spin itself is difficult to reach remotely, but the phase reference is defined by classical electronics: a reference clock, a microwave synthesizer, an arbitrary waveform generator, and the cables between them. Signal injection into the analog front ends of sensors is well documented [3], [4]. For an NV sensor the required manipulation is small: a 2.8 kHz drive offset, 0.89 ppm of the 3.15 GHz transition, imitates a 100 nT field change, and every tone derived

from a shifted reference clock inherits the clock’s fractional error, so pulling the usual 10 MHz reference by about 10 Hz, or rewriting the synthesizer’s frequency setting, is enough.

Existing work does not address this threat. NV sensing research treats control errors as benign imperfections to be averaged out by robust or double-quantum sequences [5]–[7]. Spoofing defenses for classical active sensors hide probing times from the attacker [8], a strategy that fast attackers can bypass [9]. Anomaly detectors are usually judged by detection rate, although for a field estimate what matters is whether the released value is correct after an alarm. This paper asks: if an attacker manipulates the microwave drive of an NV Ramsey magnetometer, what can the defender detect, and how much of the field estimate can hardware architecture recover? We make three contributions:

- Control-chain attack surface (Section III). A drive-frequency offset is equivalent to a field shift in the photon statistics for any schedule. Secret timing defeats only precomputed phase attacks, and dual-transition driving analytically reduces the field bias of a 1 ppm clock error from about 112 nT to 10 nT at a 10 mT bias field.
- Reference architecture and design equation (Section IV). Fallback to a reference sensor is bounded by the reference’s own error. A closed-form equation predicts the residual harm and a blind window of small spoofs, and gives the reference noise needed to close it.
- Engineering evaluation and design rules (Sections V–VI). A 30-pipeline simulation study quantifies attack success, clock sharing, and the precision, availability and cost of fallback, abstention and fusion, and maps each countermeasure to the hardware it requires.

II. Sensor Architecture and Threat Model

Fig. 1 shows the architecture. A trusted scheduler chooses interrogation times t_k and analysis phases ϕ_k and records them. The synthesizer and waveform generator turn these commands into pulses on a single NV transition, the optical readout returns photon counts, and the

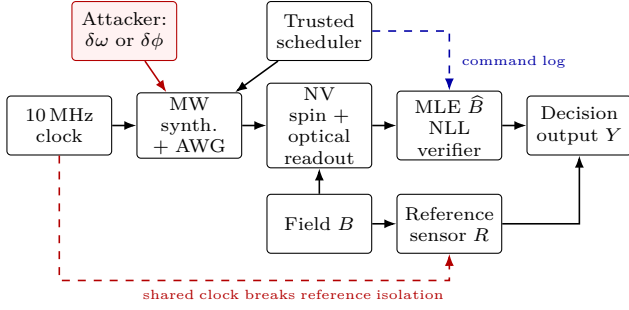


Fig. 1. Sensor chain and trust boundaries. The attacker offsets the drive frequency ($\delta\omega$) or analysis phase ($\delta\phi$) after the trusted scheduler; the command log stays protected. The reference sensor is secure only if it shares no clock, synthesizer or data path with the primary chain (red dashed path).

estimator computes a field estimate $\hat{B}$ from the counts and the logged commands. A verifier scores held-out settings, and an optional reference sensor supplies an independent reading R of the same field.

The attacker acts between the scheduler and the spin: it can add a constant drive-frequency offset or change the analysis phase of each pulse, either from a precomputed table or with real-time knowledge of the schedule. It cannot observe the true field or the reference and cannot alter counts, logs, software or the reference path; an attacker with that access controls the output directly and must be stopped by data-path protection. A release Y is harmful if $|Y - B| > \epsilon_H = 25 \text{ nT}$, and the attack targets a 100 nT offset. We also report coverage (fraction of attempts with a released output) and root-mean-square error (RMSE) over released outputs.

III. What the Primary Sensor Can Reveal

With preparation phase β , analysis phase ϕ and spin-drive detuning $\delta\omega$, the probability of reading $m_s = 0$ after a Ramsey sequence is [10], [11]

$$p = \frac{1}{2} [1 + (1 - 2\eta) c e^{-t/T_2^*} \cos \Theta], \quad \Theta = (\gamma B + \delta\omega)t + \beta - \phi, \quad (1)$$

with gyromagnetic ratio γ , contrast c , dephasing time T_2^* and symmetric readout error η ; N shots at each setting give binomial counts.

Frequency offsets are invisible. The field enters Eq. (1) only through the net detuning $\gamma B + \delta\omega$. More generally, whenever the rotating-frame dynamics of the addressed transition depend on B only through this detuning, lowering the drive frequency by $\gamma\Delta B$ produces exactly the count distribution of a field $B + \Delta B$, for any pulse shape and any schedule, including secret ones. No test on the primary data can then separate the two, and the attack is undetectable in the sense of [12]. The condition neglects the unaddressed transition, counter-rotating terms and the ^{14}N nuclear spin. A lab-frame simulation of the coupled electron-nuclear system with finite 25 ns pulses and no rotating-wave approximation gives a largest population

difference, over eight random settings per bias field, of 1.1×10^{-4} at a 10 mT bias (4.8×10^{-4} at 5 mT), against a median spoof signal of 0.034 and a per-setting shot noise of 0.016. If the residual stays at this level, even a detector that knew it exactly would need at least 5,493 sequences at 10 mT (293 at 5 mT) to detect it with 50% power at a 1% false-alarm level, so the attack is undetectable within any single sequence.

Secret timing stops precomputed phase attacks. A phase actuator imitates the field shift only if it applies $\delta\phi_k = -\gamma\Delta B t_k$ with the true interrogation time. An attacker that must commit to a table built for the nominal times t_{nom} leaves a residual phase $\gamma\Delta B (t_{\text{nom}} - t_k)$ that varies from setting to setting when the t_k are drawn secretly, and a held-out likelihood test detects the resulting misfit. A frequency offset needs no such knowledge, so secret timing does not help against it.

Dual-transition driving cancels common clock errors. With transition frequencies $\omega_{\pm} = D \pm \gamma(B_0 + B)$ and tone offsets $a_{\pm}$, the difference of the two measured detunings contains $2\gamma(B_0 + B) - (a_+ - a_-)$, so offsets common to both tones and drifts of the zero-field splitting D cancel [6], [7], [13]. A fractional clock error ϵ scales both tones; at $\epsilon = 1 \text{ ppm}$ and $B_0 = 10 \text{ mT}$ it biases a single-transition estimate by about 112 nT but a dual-transition estimate by only 10 nT. To spoof the dual-transition sensor, the attacker must inject a precise differential offset on two tones. This is an algebraic capability bound, not a demonstrated defense, and a single-transition sensor still needs an external reference.

IV. Reference-Assisted Verification

Architecture. Each sequence splits its settings into an estimation set, on which the maximum-likelihood estimate $\hat{B}$ is fitted, and a verification set, on which the negative log-likelihood score S_N is computed. A second score, $S_R = |\hat{B} - R|$, compares the primary estimate with the reference, modeled as

$$R = B + b_R + \lambda\Delta B + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_R^2), \quad (2)$$

where a bias b_R and a coupling λ describe imperfect isolation ($b_R = \lambda = 0$ for an independent reference). The alarm fires when either score exceeds its threshold; each threshold is a quantile of clean calibration scores at level $\alpha/2$, so the clean false-alarm rate targets $\alpha = 1\%$ [14]. We compare four output policies built on this combined alarm: fallback releases R on alarm and $\hat{B}$ otherwise; abstention releases $\hat{B}$ only without alarm; always-reference ignores the primary sensor; and fusion rules combine $\hat{B}$ and R continuously.

The reference sets a floor. A fallback release is harmless only if an accepted $\hat{B}$ or a substituted R is accurate. When the spoof makes $\hat{B}$ harmful, only the second case remains, so fallback harm is at least the reference's own harm, whatever the alarm rule. The same holds in the worst case for every fusion rule of the form $Y = R + g(\hat{B} - R)$, which

includes fallback, inverse-variance weighting and clipping. For a Gaussian reference and a primary error that does not depend on the field, when the attacker may choose the spoof size $\hat{B}$ carries no usable information about B , and worst-case harm cannot fall below the reference floor $2\Phi(-\epsilon_H/\sigma_R)$.

Design equation. For a spoof $d > h = \epsilon_H$ with an ideal primary estimate $\hat{B} = B + d$, a reference shift $s = b_R + \lambda d$ and discrepancy threshold κ , fallback harm is

$$H_F(d) = P(|s + \varepsilon| > h) + P(|s + \varepsilon| \leq h, |d - s - \varepsilon| \leq \kappa). \quad (3)$$

The first term is the reference floor; the second counts accurate references that fail to raise the alarm, so the spoofed $\hat{B}$ is released. For an independent reference ($s = 0$) and ignoring the likelihood alarm, harm is 1 in a blind window $h < d \leq \kappa - h$, falls in $\kappa - h < d < \kappa + h$, and reaches the floor for $d \geq \kappa + h$. With primary noise σ_P , the calibrated threshold is approximately

$$\kappa \approx z_{1-\alpha/4} \sqrt{\sigma_R^2 + \sigma_P^2}, \quad (4)$$

with $z_{1-\alpha/4} = 2.807$ for a two-sided test at level $\alpha/2$. The blind window exists when $\kappa > 2h$; at $\sigma_R = 20$ nT it spans 25 to 32.6 nT. For $\sigma_P \approx 4.28$ nT this happens once $\sigma_R > 17.3$ nT, which turns Eqs. (3) and (4) into a sizing rule for the reference sensor.

V. Evaluation

Setup. We simulate the Ramsey model of Eq. (1) with $T_2^* = 100 \mu\text{s}$, $\eta = 0.1$, five nominal times from 5 to $60 \mu\text{s}$ or secret times drawn uniformly in that range, two analysis phases, and 1000 shots per setting (20 settings per sequence). Fields are drawn from $[0, 1.9] \mu\text{T}$, the nominal reference has $\sigma_R = 20$ nT, and the nominal spoof is 100 nT. Thirty independently seeded pipelines each calibrate their own thresholds and attacker tables and evaluate 512 clean/attacked pairs per condition, 15,360 attempts in total. Policies, conditions, endpoints and seeds were fixed and time-stamped before a single evaluation run. Intervals are 95% bootstrap intervals obtained by resampling the 30 pipelines. The closed-form comparison and the fusion replay were added after the run and are reported as post-hoc analyses.

Attack capability. With a public, fixed schedule, a precomputed phase attack reaches its target undetected on 14,758 of 15,360 attempts; with secret interrogation times it succeeds on 1. A phase table optimized against the secret schedule shifts the estimate in 6,081 attempts but passes the verifier in only 17; this is a lower bound on attacker capability, since the search is finite. A drive-frequency offset succeeds undetected on 14,892 of 15,360 attempts even with secret timing, and simulated count distributions match the field-shifted ones to within 5.55×10^{-16} .

Fallback reaches the reference floor. At the nominal operating point, combined fallback releases a harmful estimate on 20.78% [20.26, 21.30] of attacks, identical

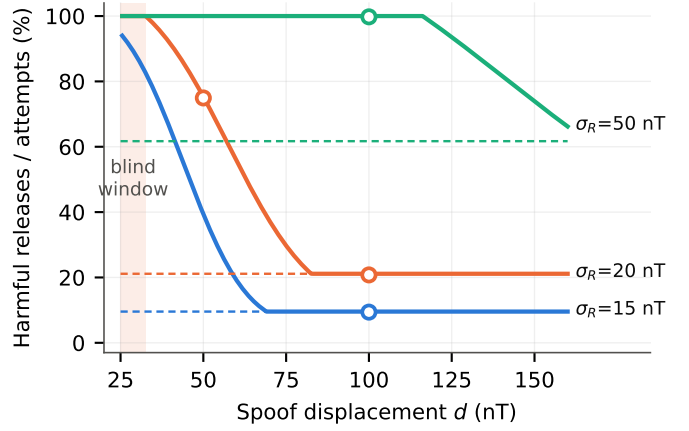


Fig. 2. Fallback harm versus spoof size predicted by Eq. (3) for three reference noise levels (lines), their reference floors (dashed) and simulated observations with 95% intervals (circles). Shading: blind window at $\sigma_R = 20$ nT, where small spoofs are released without alarm.

TABLE I
Reference dependency audit: harmful releases of combined fallback under a 100 nT spoof, $\sigma_R = 20$ nT.

Shared dependency	Model effect	Harm
None (independent)	$b_R = \lambda = 0$	20.78%
Calibration or offset error	$b_R = 10$ nT	26.94%
Partially coupled paths	$\lambda = 0.5$	89.67%
Shared clock or synthesizer	$\lambda = 1$	99.98%
Shared data path/software	attacker rewrites R	out of scope

to always-reference; this rate is the reference's own error at $\sigma_R = 20$ nT, not an effect the alarm can remove. A 50 nT spoof lies in the transition region and raises harm to 74.93%. A noisier reference with $\sigma_R = 50$ nT widens κ until the 100 nT spoof falls inside the blind window: fallback harm rises to 99.76%, worse than the reference alone (62.25%). A precise reference with $\sigma_R = 5$ nT removes harm entirely. Fig. 2 shows that Eq. (3), evaluated with the calibrated thresholds and no fitting to attack data, matches all 8 tested conditions ($\sigma_R \in \{5, 15, 20, 50\}$ nT, a 50 nT spoof, a 10 nT reference bias, and $\lambda \in \{0.5, 1\}$) within 0.60 percentage points.

Shared hardware defeats the reference. Table I maps shared dependencies to Eq. (2). A 10 nT reference offset raises harm to 26.94%; partial coupling ($\lambda = 0.5$) raises it to 89.67%. A reference that inherits the full spoof, as a second NV sensor on the same clock would, releases harmful estimates on 99.98% of attacks, while alarms fire on only 0.94% because both channels move together.

Precision, availability and cost. Table II compares output rules replayed on the stored per-attempt records. Fallback's only advantage over always-reference is clean precision: RMSE falls from 19.98 to 6.14 nT (paired difference -13.84 nT, interval [-14.27, -13.44] adjusted for multiple comparisons), bought with 20,000 primary shots per estimate. Inverse-variance fusion is the most precise

TABLE II

Output rules replayed on the frozen per-attempt records, $\sigma_R = 20$ nT. Gated: reference on alarm, else inverse-variance fusion. Clipped: $R + \text{clip}(\hat{B} - R, \pm\kappa)$. Worst case: model maximum over spoofs of 0–300 nT.

Rule	Clean RMSE (nT)	Attacked 100 nT	harm (%) 50 nT	Worst case (%)
Always reference	19.98	20.78	20.78	21.2
Combined fallback	6.14	20.78	74.93	96.1
Gated fusion	6.09	20.78	74.93	95.4
Inverse-variance fusion	4.29	100.00	100.00	100.0
Clipped fusion	4.38	94.86	94.86	94.9

rule on clean data, but it weights the compromised primary sensor heavily and releases a harmful estimate on every attack. Every rule other than always-reference has a worst case far above the reference floor, which is only a lower bound. Abstention lowers harm by 18.85 percentage points only by releasing almost nothing: coverage under attack drops by 98.07 points, all 296 released outputs are harmful, and each release costs 1.04 million primary shots. Clean false alarms occur at 1.07% [0.91, 1.24], close to the 1% target, and a genuine 100 nT field change raises only 154 alarms in 15,360 attempts because both sensors see it. Finally, if reference errors are independent from reading to reading, averaging 11 readings gives $\sigma_R = 6.03$ nT, matching fallback’s clean precision with a floor of 0.003%, so fallback is worthwhile only when primary shots are much cheaper than reference readings or low-frequency drift prevents averaging.

VI. Design Rules and Limitations

Table III maps each countermeasure to what it stops and the hardware it needs; the results reduce to four rules for sensor designers. First, match the defense to the actuator: secret timing costs only scheduler logic and stops precomputed phase attacks, but not frequency offsets. Second, size the reference to the application’s tolerance, not to the detector’s alarm rate: for a 25 nT tolerance the reference noise must stay below 17.3 nT, or small spoofs pass through the blind window. Third, audit the reference for shared clocks, synthesizers and data paths, since a shared clock turns the reference into a second victim. Fourth, prefer drive schemes that address both spin transitions, which cancel common clock errors at the cost of a second tone. Integrity monitoring in satellite navigation follows the same logic of separating fault detection from fault exclusion [15], [16].

All results are from simulation. The reference is an abstract Gaussian channel without drift, bandwidth or alignment effects; the attacker is modeled at the level of frequency and phase offsets rather than as a circuit-level synthesizer or clock model; the attacker’s phase-table search is finite; and the 30 pipelines replicate sampling and calibration variance within one code base.

TABLE III

Countermeasures, what they stop, and their hardware cost (from Sections III–V).

Countermeasure	Stops	Hardware cost
Secret interrogation timing	Precomputed phase attacks (1 of 15,360 succeed); not frequency offsets	Randomized scheduler; no added hardware
Dual-transition drive	Common clock/synthesizer errors (112 to 10 nT per ppm, analytic)	Second tone: extra synthesizer channel or IQ sideband
Independent reference with fallback	Frequency offsets, down to the reference’s own error rate	Second sensor, own clock, $\sigma_R < 17.3$ nT for a 25 nT tolerance
Dependency audit	Shared-clock failure (99.98% harm otherwise)	Separate clock, synthesizer and data path

VII. Conclusion

A single-transition NV Ramsey magnetometer cannot tell a sub-ppm microwave drive offset from a real field change, so its integrity depends on hardware outside the spin. An independent reference can detect the attack, but under attack its own error rate is the best that fallback can achieve, and a noisy reference opens a blind window in which small spoofs pass unnoticed. The closed-form design equation, which matches the simulation, lets designers size the reference to the application’s tolerance, and a dependency audit shows that sharing a clock with the primary sensor removes the protection entirely. Future work will simulate dual-transition sensors under two-tone attacks, model the clock and synthesizer at circuit level, and demonstrate drive spoofing and reference-assisted verification on NV hardware.

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